Sharp vanishing order for biharmonic equations with bounded potentials

Abstract. The exponent $1/3$ in normalized vanishing-order bounds for biharmonic equations with bounded potentials is optimal in every dimension $n \geq 2$, even when both the solution and the potential are real. For each sufficiently large even integer N, we construct a bounded real solution of $\Delta^2 u_N + V_N u_N = 0$ on the plane with $\operatorname{ord}_0 u_N = N + 2$, $\|V_N\|_\infty \leq CN^3$, and $\|u_N\|_{L^\infty(B_1)} = \|u_N\|_{L^\infty(\mathbb{R}^2)} = 1$. The construction joins biharmonic modes across annuli and removes the residual near the real zeros by clamped replacement on sparse sets. Signed radial averaging with vanishing second moment preserves the replacement near every zero away from the origin, without a maximum principle. The solution is globally C^4, and both the solution and the potential are smooth away from the origin. The same family also establishes sharpness in the complex-valued classes.

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1 Introduction and main results

We study the dependence of vanishing order on the size of a bounded potential in the biharmonic equation

$$\Delta^2 u + Vu = 0, \qquad \Delta = \sum_{j=1}^{n} \partial_j^2. \tag{1.1}$$

A bound in terms of the potential alone requires a normalization of the solution. Here the domain and two comparison scales are fixed, and the ratio of the outer and inner supremum norms is controlled independently of the potential. Under this normalization, we determine the smallest power of $\|V\|_\infty$ in a uniform bound for the vanishing order. The exponent is $1/3$ in every dimension $n \geq 2$. The examples proving sharpness have both real solutions and real potentials; hence the same exponent is optimal in the real, real-potential, and fully complex classes.

1.1 Background and related results

The question belongs to quantitative unique continuation. Strong unique continuation excludes zeros of infinite order, whereas a quantitative estimate controls the finite order in terms of the coefficients and an appropriate normalization. The second-order theory provides two complementary approaches: singular-weight Carleman inequalities, as in Jerison and Kenig [14], and frequency-function arguments, developed by Garofalo and Lin [10, 11]. Both approaches relate local vanishing to estimates comparing the size of a solution at different scales. The normalization is part of such an estimate, rather than a choice that can be made after the constants have been fixed.

Spectral problems suggest a natural scale for this question. For a nonzero eigenfunction of $-\Delta_g$ on a fixed smooth, closed, connected Riemannian manifold, Donnelly and Fefferman [7] bounded the vanishing order by a constant multiple of the square root of the eigenvalue. Kukavica [17] studied the corresponding question for variable potentials. For bounded potentials in the second-order equation, the quantitative estimates of Bourgain and Kenig [3], arising in Anderson localization, lead instead to the power $2/3$. Kenig [15, p. XX–4] formulated the normalized local problem explicitly and asked whether the power could be reduced to $1/2$ for real solutions and real potentials.

The distinction between these powers reflects both the coefficient class and the role of complex values. Meshkov [23] constructed complex solutions and bounded complex potentials with decay of order $\exp(-c|x|^{4/3})$ at infinity. Bakri [2] proved quantitative vanishing estimates on closed manifolds and gave a complex construction showing the sharpness of the bounded-potential power $2/3$. Under differentiability assumptions on the potential, stronger bounds involve its derivatives: see Bakri [1] and the weighted-frequency approach of Zhu [25]. Estimates for singular lower-order coefficients were developed further by Davey and Zhu [6] and Davey [5]. These results distinguish bounds controlled only by a potential norm from bounds that also depend on additional regularity.

For powers of the Laplacian, strong unique continuation with singular lower-order coefficients was studied by Colombini and Grammatico [4] and Lin [18]. Lin, Nagayasu and Wang [19] obtained quantitative estimates through three-sphere and doubling inequalities. Higher-order frequency methods were considered in [25], and Zhu [26] established quantitative estimates with coefficients in Lebesgue spaces. In the biharmonic case, the iterated Carleman inequality of Huang, Wang and Zheng [13, Lemma 2.1] has a factor λ^6 on the left-hand side. It absorbs a bounded zeroth-order

term once $\lambda \geq C(1 + \|V\|_\infty^{1/3})$, which identifies the bounded-potential exponent relevant here. Their separate sharpness example concerns a different fourth-order constant-coefficient operator, not Δ^2.

Teng, Wang and Zhu [24, Theorem 2] obtain the same bounded-potential power $1/3$ by weighted frequency functions, with normalization on two fixed balls. We prove the upper bound for the precise class used here in Section 7, directly from the Carleman inequality in [13]; thus the upper estimate is independent of the construction. It applies to complex potentials and requires no derivatives of V. This last point matters when comparing with stronger-regularity results. Under their normalization, Liu, Tian and Yang [21, Theorem 1.1] assume $V \in W^{1,\infty}$ and obtain a bound involving both $\|V\|_\infty^{1/4}$ and $\|\nabla V\|_\infty$. Such a bound does not yield a $1/4$-power estimate in the L^∞ norm alone. Likewise, the biharmonic spectral problems treated by Lin and Zhu [20] have specified boundary conditions and a spectral parameter; they are not the local problem with an arbitrary bounded potential considered here.

Reality presents a further issue, separate from regularity. In the second-order planar problem, Kenig, Silvestre and Wang [16] proved quantitative Landis estimates under a sign condition on the potential. Logunov, Malinnikova, Nadirashvili and Nazarov [22] removed the sign condition in their planar result on exponential decay, using specifically two-dimensional methods. In contrast, Frank and Ivanisvili [9] constructed real counterexamples with $4/3$-power exponential decay in dimensions at least three. Feng, Gan, Jiang and Lin [8, Theorem 1.1] constructed real Dirichlet solutions in a fixed three-dimensional ball with vanishing order k and potential norm at most $Ck^{3/2}$. These results concern different normalizations and, in the global case, additional control at infinity. They motivate the real-valued construction but are not substitutes for a sharpness example in the fixed two-scale class below.

The obstruction to obtaining a real example by taking real parts is already visible in the equation. If $u = u_1 + iu_2$ and $V = V_1 + iV_2$, then $\Delta^2 u_1 + V_1 u_1 = V_2 u_2$. Thus u_1 need not solve a homogeneous scalar equation with a bounded real potential. Taking real parts also introduces zeros at which the quotient defining the potential may become singular. We instead build a real approximate solution and remove its residual near the entire small-value set. Sparse-set replacement and variable-radius averaging follow the general strategy of [9, Sections 6–9] and [8, Section 4], but the fourth-order correction requires a different inverse estimate and a different averaging kernel. Positivity is not available as a general principle for clamped biharmonic problems; see [12]. Our inverse is obtained from a planar H^2 estimate, and the averaging kernel is chosen to reproduce biharmonic functions exactly.

1.2 The normalized classes

Write $B_r^n(p)$ for an open Euclidean ball and $B_r^n = B_r^n(0)$. All supremum norms are essential suprema. For a locally bounded function, set

$$\operatorname{ord}_p u = \sup\{\gamma \geq 0 : \|u\|_{L^\infty(B_r(p))} = O(r^\gamma) \text{ as } r \downarrow 0\}. \tag{1.2}$$

The implied constant may depend on the individual function and on γ. The zero need not be isolated.

Definition 1.1. For $n \geq 2$, $C_0 \geq 1$ and $M \geq 1$, let $\mathcal{A}_n^{\mathbb{C}}(C_0, M)$ consist of the pairs satisfying

$$\begin{aligned} \Delta^2 u + Vu = 0 \quad &\text{in } \mathcal{D}'(B_{10}^n), \\ V \in L^\infty(B_{10}^n; \mathbb{C}), \qquad &\|V\|_\infty \leq M, \\ u \in H^4_{\mathrm{loc}}(B_{10}^n; \mathbb{C}) \cap L^\infty(B_{10}^n), & \\ \|u\|_{L^\infty(B_{10}^n)} \leq C_0, \qquad &\|u\|_{L^\infty(B_1^n)} \geq 1. \end{aligned} \tag{1.3}$$

The subclass with real u, V is denoted by $\mathcal{A}_n^{\mathbb{R}}$; that with real V and possibly complex u is denoted by $\mathcal{A}_n^{\mathbb{R},\mathbb{C}}$. No boundary condition or sign restriction on a real potential is imposed. For each class, let α_n with the corresponding superscript be the infimum of all $\alpha \geq 0$ such that, for every $C_0 \geq 1$, there is $C = C(n, C_0, \alpha)$ for which

$$\operatorname{ord}_p u \leq C(1 + M^\alpha), \qquad p \in B_{1/2}^n, \tag{1.4}$$

holds for every $M \geq 1$ and every pair in that class.

The two-scale condition cannot be omitted. The harmonic polynomials $10^{-N} \operatorname{Re}(x_1 + ix_2)^N$ have outer supremum one and arbitrarily large vanishing order with zero potential, but do not satisfy the inner normalization.

1.3 Main results

The main construction is planar and entirely real. The common upper bound then identifies the optimal exponent in each normalized class.

Theorem 1.2 (Real examples)**.** *There are absolute constants $C > 0$ and N_0 such that, for every even integer $N \geq N_0$, there are real functions*

$$u_N \in C^4(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2), \qquad V_N \in L^\infty(\mathbb{R}^2)$$

satisfying, pointwise and distributionally,

$$\Delta^2 u_N + V_N u_N = 0, \qquad \|V_N\|_\infty \leq CN^3, \qquad \operatorname{ord}_0 u_N = N + 2. \tag{1.5}$$

Moreover,

$$\|u_N\|_{L^\infty(\mathbb{R}^2)} = \|u_N\|_{L^\infty(B_{10}^2)} = \|u_N\|_{L^\infty(B_1^2)} = 1. \tag{1.6}$$

Both functions are smooth on $\mathbb{R}^2 \setminus \{0\}$. With $z = x_1 + ix_2$, there is a nonzero real d_N such that

$$D^\beta(u_N - d_N |z|^2 \operatorname{Re}(z^N)) = O_N(|z|^{N+6-|\beta|}), \qquad |\beta| \leq 4. \tag{1.7}$$

Proposition 1.3 (The common upper bound)**.** *For every $n \geq 2$ and $C_0 \geq 1$, there are $c, C > 0$, depending only on n, C_0, such that every pair in $\mathcal{A}_n^{\mathbb{C}}(C_0, M)$ satisfies*

$$\|u\|_{L^\infty(B_r^n(p))} \geq c r^{C(1+M^{1/3})}, \qquad p \in B_1^n, \quad 0 < r < 1/4. \tag{1.8}$$

In particular, $\operatorname{ord}_p u \leq C(1 + M^{1/3})$ *on* $B_{1/2}^n$.

Corollary 1.4. *For every $n \geq 2$,*

$$\alpha_n^{\mathbb{R}} = \alpha_n^{\mathbb{R},\mathbb{C}} = \alpha_n^{\mathbb{C}} = \frac{1}{3},$$

and the estimate is valid at the endpoint $\alpha = 1/3$ in all three classes. The same real family, extended constantly in the last $n - 2$ coordinates, proves sharpness in each class.

The theorem supplies the lower obstruction in the smallest of the three classes, while Proposition 1.3 gives the upper estimate in the largest. Constant extension preserves the equation, the normalization, and the vanishing order. No separate complex-valued construction is therefore needed. The assertion is sharpness of the power in a uniform estimate, not of its multiplicative constant.

These examples rule out smaller exponents, but do not contradict strong unique continuation: each zero has finite order, and the potential norm grows with that order. Neither global smoothness nor compact support of V_N is asserted. The result concerns $\Delta^2 + V$ with bounded zeroth-order potentials and fixed two-scale normalization; it does not assert the same exponent for arbitrary fourth-order elliptic operators. The distinction from prescribed decay at infinity is discussed in Section 8.

1.4 Proof mechanism

The construction begins with annular transitions from frequency ℓ to $\ell+2k$, with residual bounded by $C\ell^2k^2r^{-4}$ relative to a positive envelope. The two radial branches $r^{-p}\cos(p\theta)$ and $r^{2-p}\cos(p\theta)$ allow the profile to remain exactly biharmonic throughout the region of possible amplitude cancellation. A finite frequency schedule ensures $\ell k \leq N^{3/2}$ and keeps $\sum k^{-1}$ bounded, so that the transitions fit in a fixed annulus. Kelvin inversion then produces the prescribed polynomial leading term at the origin.

To control the quotient near the real zeros, we correct the resulting function f_N on $D_N = \{|f_N| < \tau E_N\}$. A uniform density estimate makes D_N sparse on balls of radius μ_N^{-1}. The clamped inverse gives a single correction satisfying

$$|w_N| \leq C\nu^2\tau E_N, \qquad |w_N| \leq C\nu^2 N^3 \mu_N^{-4} E_N.$$

The first estimate keeps the smoothed function away from zero outside the smaller small-value region contained in D_N; the second preserves the leading polynomial and controls the fourth-order residual after smoothing. The averaging kernel has vanishing second moment, so it reproduces the biharmonic replacement near every zero away from the origin. On the remaining region, the lower bound relative to the envelope controls the quotient potential.

There are two distinct cancellations in this procedure. The radial branches eliminate the residual where their amplitudes can cancel; the zero second moment eliminates the Laplacian term in the

radial mean-value formula for a biharmonic function. The latter kernel is necessarily signed. Its purpose is exact reproduction, not positivity. The clamped correction is solved on the sparse open set and extended by zero as an H^2 function. No equation for the fourth derivatives of that extension across the boundary is assumed.

The proof follows these dependencies. Section 2 records the radial and Kelvin identities. Section 3 constructs the approximate solution and its positive envelope, and Section 4 proves the uniform small-value density estimate. The inverse and averaging lemmas are established in Section 5 and combined in Section 6 to produce the bounded real potential. Section 7 proves the upper bound and the passage to higher dimensions independently of the construction.

2 Preliminaries

Throughout the construction, $\Omega = \mathbb{R}^2 \setminus \{0\}$. We use $D^j f$ for the tensor of all ordered Cartesian derivatives of order j, and $\|D^2 f\|_2^2 = \sum_{i,j} \|\partial_{ij} f\|_2^2$. For an arbitrary open set $D \subset \mathbb{R}^2$, $H_0^2(D)$ is the closure of $C_c^\infty(D)$ in $H^2(D)$. Zero extension maps this space into $H^2(\mathbb{R}^2)$, as follows by approximation. This notation imposes no regularity on ∂D. On a smooth boundary, the clamped traces are $w = \partial_\nu w = 0$.

Unless otherwise indicated, the constants C, C_j in the construction depend only on the fixed smooth profiles and on j, and are uniform in the frequencies and radii. All auxiliary parameters are fixed before N is allowed to vary; the constants in (1.7) may depend on N.

Lemma 2.1 (Radial identities)**.** *With $t = \log r$ and $D = \partial_t$,*

$$\Delta^2 = e^{-4t}((\partial_t - 2)^2 + \partial_\theta^2)(\partial_t^2 + \partial_\theta^2). \tag{2.1}$$

For an integer $m \geq 1$,

$$\Delta^2[r^{-m} e^{im\theta} b(t)] = r^{-m-4} e^{im\theta} P_m(D) b, \tag{2.2}$$

where

$$\begin{aligned} P_m(D) &= D(D-2)(D-2m)(D-2m-2) \\ &= D^4 - (4m+4)D^3 + (4m^2+12m+4)D^2 - (8m^2+8m)D. \end{aligned}$$

Consequently, for $1 \leq k \leq m$,

$$|P_m(k\partial_x) b| \leq C m^2 k^2 \max_{1 \leq j \leq 4} \|b^{(j)}\|_\infty. \tag{2.3}$$

Also, for real s,

$$\Delta^2(r^s e^{im\theta}) = (s^2 - m^2)((s-2)^2 - m^2) r^{s-4} e^{im\theta}. \tag{2.4}$$

Proof. The identity $\Delta = e^{-2t}(\partial_t^2 + \partial_\theta^2)$ gives (2.1). Conjugating by e^{-mt} replaces ∂_t by $D - m$, which yields (2.2). The derivative terms are bounded using $k^4 + mk^3 + m^2k^2 + m^2k \leq Cm^2k^2$. Applying the two factors in (2.1) to $e^{st+im\theta}$ gives (2.4). □

Lemma 2.2 (Kelvin transformation)**.** *For a smooth function v on a planar annulus, set $Kv(r,\theta) = r^2 v(1/r,\theta)$. Then*

$$\Delta^2(Kv)(r,\theta) = r^{-6}(\Delta^2 v)(1/r,\theta). \tag{2.5}$$

Proof. Write $g(s,\theta) = v(e^s,\theta)$ and $s = -t$. Applying (2.1) to $e^{2t}g(-t,\theta)$ gives e^{-2t} times the commuting factors $\partial_s^2 + \partial_\theta^2$ and $(\partial_s - 2)^2 + \partial_\theta^2$ acting on g. The corresponding expression for $(\Delta^2 v)(e^{-t},\theta)$ has the prefactor e^{4t}. Comparing the two expressions proves the identity. □

3 Construction of the approximate solution

We construct annular profiles with controlled residuals, then choose a frequency schedule that keeps all transitions in a fixed annulus.

3.1 Annular transitions

Fix $\delta = 10^{-3}$ and real smooth cutoffs with values in $[0,1]$:

	equals 0	equals 1
$\chi_2(x)$	$x \le -\delta$	$x \ge \delta$
$\chi_1(x)$	$x \ge 3/4$	$x \le 1/4$
$\eta(x)$	$x \le 1$	$x \ge 7/4$.

For integers $\ell \ge 16$, $1 \le k \le \ell$, and $\rho > 0$, put

$$p = \ell + 2k, \quad r_c = \rho e^{1/k}, \quad r_+ = \rho e^{3/k}, \quad x = k\log(r/r_c) \in [-1,2]. \tag{3.1}$$

Set

$$L_k(x) = 20k(e^{2x/k} - 1), \qquad H_k(x) = (1-\eta(x))L_k(x) + \eta(x)L_k(2), \qquad Q_k(x) = e^{-2x}L_k(x).$$

Define

$$F_{\ell,k,\rho} = \chi_1(x) r^{-\ell}\cos(\ell\theta) - \chi_2(x) r_c^{2k} r^{-p} H_k(x)\cos(p\theta). \tag{3.2}$$

Write

$$A = r^{-\ell}, \quad b_1 = \chi_1, \quad b_2 = \chi_2 e^{-2x}H_k, \quad Z = b_1 - b_2 e^{2ik\theta}, \quad a = k/\ell.$$

Choose $\omega \in C_c^\infty((0.002, 0.2))$, with $0 \le \omega \le 1$ and $\omega = 1$ on $[0.005, 0.15]$, and set

$$E_{\ell,k,\rho} = A(|Z|^2 + a^2\omega(x)^2)^{1/2}. \tag{3.3}$$

Thus $F_{\ell,k,\rho} = A\,\mathrm{Re}(e^{i\ell\theta}Z)$. The complex amplitude is used only as notation; both the profile and the envelope are real.

Lemma 3.1 (Profiles and envelopes)**.** *The envelope is positive and smooth. Uniformly in ℓ, k, ρ,*

$$|F_{\ell,k,\rho}| \le E_{\ell,k,\rho} \le C_A r^{-\ell}, \tag{3.4}$$

$$|D^j F_{\ell,k,\rho}| + |D^j E_{\ell,k,\rho}| \le C_j (\ell/r)^j E_{\ell,k,\rho}, \qquad j \ge 1, \tag{3.5}$$

$$|\Delta^2 F_{\ell,k,\rho}| \le C\ell^2 k^2 r^{-4} E_{\ell,k,\rho}. \tag{3.6}$$

The profile is biharmonic throughout $\delta < x < 1/4$. On collars of the inner and outer boundaries, respectively, the profile and envelope are

$$(r^{-\ell}\cos\ell\theta, r^{-\ell}), \qquad (-r_c^{2k} L_k(2) r^{-p}\cos p\theta, r_c^{2k} L_k(2) r^{-p}).$$

The ratio of the outer to the inner envelope values is at most $80e^{-27/4} < 1$.

Proof. The derivatives of L_k, H_k, Q_k of each fixed order are uniformly bounded on $[-1, 2]$. For $j \ge 1$, this follows from $L_k^{(j)} = 20\, 2^j k^{1-j} e^{2x/k}$; for order zero, use $|L_k(x)| \le 40|x|e^{2|x|}$. The identity

$$r_c^{2k} r^{-p} L_k(x) = 20k r_c^{2k} (r_c^{-2} r^{2-p} - r^{-p}) \tag{3.7}$$

together with (2.4) shows that the profile is exactly biharmonic in the crossing region, where $\chi_1 = \chi_2 = 1$ and $\eta = 0$.

Amplitude cancellation does not occur where the profile cutoffs vary:

$$|Q_k(x)| \le 40\delta e^{4\delta} < 1/4 \quad (|x| \le \delta), \qquad Q_k(x) \ge 40xe^{-2x} > 5 \quad (1/4 \le x \le 3/4).$$

For the second inequality, the only interior critical point of xe^{-2x} is a maximum, and both endpoint values exceed $1/8$. On $1 \le x \le 2$, we also have $H_k \ge L_k(1) \ge 40$. Hence $|Z| \ge c > 0$ wherever a profile cutoff varies. Applying (2.3) there at frequencies ℓ and $p \le 3\ell$ bounds the residual by $C\ell^2 k^2 r^{-4} A \le C\ell^2 k^2 r^{-4} E$. The profile is exactly biharmonic in the remaining regions.

A zero of Z requires $Q_k = 1$ and $e^{2ik\theta} = 1$ in $\delta < x < 1/4$. The representation

$$Q_k(x) = 40x \int_0^1 e^{-2x(1-t/k)}\, dt \tag{3.8}$$

gives $40xe^{-2x} \le Q_k(x) \le 40x$ for $x > 0$. Hence

$$1/2 \le Q_k(x) \le 2, \quad 0 < x < 1/4 \quad \Longrightarrow \quad 1/80 \le x < 1/10. \tag{3.9}$$

Indeed, $x \ge 1/10$ would give $Q_k \ge 4e^{-1/5} > 2$. Moreover,

$$Q_k'(x) = 40e^{-2x}(k - (k-1)e^{2x/k}), \qquad 16 \le Q_k'(x) \le 40 \quad (0.005 \le x \le 0.15). \tag{3.10}$$

For the lower bound, use $k - (k-1)e^{2x/k} \ge 1 - 2xe^{2x}$ and $e^{-2x} \ge 1 - 2x$; the upper bound follows from $k - (k-1)e^{2x/k} \le 1$. The same formula shows $Q_k' > 0$ on $(0, 1/4)$, since for $k > 1$, $\log(1 - 1/k) + 2x/k < 0$; the case $k = 1$ is direct. Thus $\omega = 1$ wherever $|Z|$ can approach zero, which proves that the envelope is positive.

Where $|Z|$ is bounded below, (3.5) follows by differentiation. It remains to treat the cancellation band. Write $E = AB$ with $B = (|Z|^2 + a^2)^{1/2} \geq a$. Logarithmic-polar derivatives of Z of order $h \geq 1$ are bounded by $C_h k^h$. Derivatives of the square root give terms $C_j k^j B^{1-h}$, $1 \leq h \leq j$, and

$$\frac{k^j B^{1-h}}{\ell^j B} = \frac{a^j}{B^h} \leq a^{j-h} \leq 1.$$

For $F = A\,\mathrm{Re}(e^{i\ell\theta} Z)$ use $|Z| \leq B$ and $k^h \leq \ell^h a \leq \ell^h B$. Passing to Cartesian derivatives introduces r^{-j} and terms of lower logarithmic-polar order, proving (3.5). The cutoff ω varies only where $|Z| \geq c$, so it introduces no further loss.

The size bound and the collar formulas follow from the definitions. The ratio of the outer to the inner envelope values is

$$L_k(2) e^{-4-3\ell/k} \leq 80 e^{-4-(3\ell-4)/k} \leq 80 e^{-7+4/\ell} \leq 80 e^{-27/4},$$

where $L_k(2) \leq 80 e^{4/k}$. □

3.2 Frequency budget and assembly

The frequency increments must control both the residual and the total radial width. Let $0 < \varepsilon \leq 1/2$ and let $L \geq \max\{16, 4\varepsilon^{-2}\}$ be even. For an even $N > L$, start at $\ell_0 = L$, $\rho_0 = 2$ and set

$$\begin{aligned} k_j &= \min\{\lfloor \varepsilon \ell_j \rfloor, \lfloor N^{3/2}/\ell_j \rfloor, (N - \ell_j)/2\}, \\ \ell_{j+1} &= \ell_j + 2k_j, \qquad \rho_{j+1} = \rho_j e^{3/k_j}, \end{aligned} \tag{3.11}$$

until $\ell_J = N$.

Lemma 3.2. *The schedule terminates, and*

$$1 \leq k_j \leq \varepsilon \ell_j, \quad \ell_j^2 k_j^2 \leq N^3, \quad \sum_{j<J} k_j^{-1} < 3, \quad 2 \leq \rho_j \leq R_* := 2e^9. \tag{3.12}$$

The radius bound is independent of N, L, ε.

Proof. Before termination, every candidate for k_j is a positive integer: $\varepsilon L \geq 2$ and $N^{3/2}/\ell_j \geq \sqrt{N}$, while the remaining gap is positive and even. Thus ℓ_j reaches N after finitely many steps. The first two estimates follow from the definition.

A step strictly limited by the remaining gap is terminal and contributes at most one to $\sum k_j^{-1}$. Among the other steps, those with $k_j = \lfloor \varepsilon \ell_j \rfloor$ occur first, since this candidate increases while the other decreases. For these steps, $k_j \geq \varepsilon \ell_j / 2$ and $\ell_{j+1} \geq (1 + \varepsilon)\ell_j$, so

$$\sum \frac{1}{k_j} \leq \frac{2}{\varepsilon L} \sum_{h \geq 0} (1 + \varepsilon)^{-h} = \frac{2(1 + \varepsilon)}{\varepsilon^2 L} \leq \frac{3}{4}.$$

For the remaining steps not limited by the final gap, $k_j \geq N^{3/2}/(2\ell_j)$ and

$$\frac{1}{k_j} = \frac{\ell_{j+1} - \ell_j}{2k_j^2} \leq \frac{2}{N^3} \int_{\ell_j}^{\ell_{j+1}} t^2 \, dt.$$

Their sum is at most $2/3$. Hence the total is less than three, and the radius recurrence gives (3.12). □

Set $\gamma_0 = 2^L$ and $\gamma_{j+1} = -\gamma_j r_{c,j}^{2k_j} L_{k_j}(2)$, where $r_{c,j} = \rho_j e^{1/k_j}$. On the jth annulus use $\gamma_j F_{\ell_j,k_j,\rho_j}$ and $|\gamma_j| E_{\ell_j,k_j,\rho_j}$. The collars agree with the same harmonic expression on both sides of each interface, so the assembled functions match to all orders. Continue by

$$\gamma_J r^{-N} \cos N\theta, \qquad |\gamma_J| r^{-N} \quad (r \geq \rho_J),$$

and denote the assembled pair by $v, \mathcal{E}$. The envelope equals one at the initial boundary. The boundary-ratio estimate therefore gives

$$|v| \leq \mathcal{E} \leq C_A \qquad (r \geq 2). \tag{3.13}$$

Every γ_j is nonzero.

3.3 Inversion, outer extension, and local frequency

Kelvin inversion converts the terminal mode into the prescribed polynomial at the origin. For $0 < r \leq 1/2$, set

$$f_N(r,\theta) = r^2 v(1/r,\theta), \qquad E_N(r,\theta) = r^2 \mathcal{E}(1/r,\theta). \tag{3.14}$$

In the fixed core $0 < r < R_*^{-1}$, the formulas reduce to

$$f_N = c_N r^{N+2} \cos N\theta, \qquad E_N = |c_N| r^{N+2}, \qquad c_N = \gamma_J \neq 0. \tag{3.15}$$

Choose a fixed smooth $\zeta \in [0,1]$, zero for $r \leq 5/8$ and one for $r \geq 3/4$. For $r \geq 1/2$, extend by

$$\begin{aligned} A_L(r) &= 2^L r^{L+2} \exp[-(2L+2)\zeta(r)\log(2r)], \\ f_N(r,\theta) &= A_L(r)\cos L\theta, \qquad E_N(r,\theta) = A_L(r). \end{aligned} \tag{3.16}$$

The definitions agree on a collar of $r = 1/2$, and $A_L(r) = 2^{-L-2} r^{-L}$ for $r \geq 3/4$. Writing $\mathcal{L}_L A = A'' + r^{-1}A' - L^2 r^{-2} A$, we have

$$\Delta^2 f_N = q_L f_N, \qquad q_L = \frac{\mathcal{L}_L^2 A_L}{A_L}, \qquad |q_L| \leq C(L/r)^4, \qquad \operatorname{supp} q_L \subset [5/8, 3/4]. \tag{3.17}$$

For the bound, differentiation of $\log A_L$ on the fixed transition interval gives $|\partial_r^j \log A_L| \leq C_j L r^{-j}$ for $j \geq 1$. The product rule then yields $|A_L^{(j)}| \leq C_j (L/r)^j A_L$ for $j \leq 4$; substitution into $\mathcal{L}_L^2$ proves (3.17). Outside this interval, the two exact powers are biharmonic. Only the positive amplitude A_L appears in the denominator; no division by the cosine factor is needed.

Fix a smooth $\vartheta \in [0,1]$ with $\vartheta = 0$ on $x \leq -1/2$ and $\vartheta = 1$ on $x \geq 7/4$. On each exterior annulus define $m(r) = \ell_j + 2k_j \vartheta(x)$, with $m = N$ on the terminal tail. Set

$$\mu_N(r) = m(1/r)/r \quad (r \leq 1/2), \qquad \mu_N(r) = L/r \quad (r \geq 1/2). \tag{3.18}$$

The collar identities ensure that this defines a smooth positive function on Ω. In exterior logarithmic radius, $\partial_t^h m = O_h(k^{h+1})$ for $h \geq 1$. In particular,

$$\left| \frac{d}{dr} \frac{r}{m(1/r)} \right| \leq L^{-1} + C(k/\ell)^2.$$

Thus fixed $\varepsilon_0 > 0$ and $L_0 \geq 16$ can be chosen so that, whenever $\varepsilon \leq \varepsilon_0$ and $L \geq L_0$,

$$\mu_N \geq L/r, \qquad |\nabla \mu_N^{-1}| \leq 1/100, \qquad |D^j \mu_N| \leq C_j \mu_N^{j+1} \quad (j \geq 1). \tag{3.19}$$

For higher derivatives, the preceding bounds for m give $|\partial_t^h m| \leq C_h m^{h+1}$; conversion to radial derivatives, together with $m \geq L$, proves the last estimate. Decrease ε_0 so that every ball of radius $3/\mu_N$ meeting an assembly interface lies in the matching pure-mode collars, where the angular frequency in μ_N is constant. This choice is uniform: the collars have logarithmic width at least c/k, whereas the logarithmic width of such a ball is at most C/ℓ. The same conclusion holds after inversion.

The following estimates will be used in the density argument and the clamped correction.

Proposition 3.3 (Prepared estimates)**.** *Assume the schedule hypotheses, $\varepsilon \leq \varepsilon_0$, $L \geq L_0$, and $N^3 \geq L^4$. With constants independent of N, L, ε,*

$$|f_N| \leq E_N, \qquad |D^j f_N| + |D^j E_N| \leq C_j \mu_N^j E_N \quad (j \geq 1), \tag{3.20}$$

$$|\Delta^2 f_N| \leq M_N E_N, \qquad M_N = C_* N^3. \tag{3.21}$$

For $A = E_N$ and $A = E_N \mu_N^{-4}$ there is one fixed $K \geq 1$ such that

$$K^{-1} \leq \frac{\mu_N(y)}{\mu_N(x)} \leq K, \qquad K^{-1} \leq \frac{A(y)}{A(x)} \leq K \quad \text{if } |y - x| < 4/\mu_N(x). \tag{3.22}$$

These balls are relatively compact in Ω. For $0 < \tau < 1$, on $D_N(\tau) = \{|f_N| < \tau E_N\} \subset \Omega$,

$$|\Delta^2 f_N| \leq C(\varepsilon^2 + \tau) \mu_N^4 E_N. \tag{3.23}$$

Proof. The jet estimates follow from (3.5), the derivatives of inversion, and (3.16). In an inverted annulus, (2.5) and (3.6) give

$$|\Delta^2 f_N| \leq C \ell_j^2 k_j^2 r^{-4} E_N \leq C R_*^4 N^3 E_N, \qquad |\Delta^2 f_N| \leq C \varepsilon^2 \mu_N^4 E_N.$$

Here $R_*^{-1} \leq r \leq 1/2$ on every transition. The residual vanishes in the core. On the outer transition, (3.17) and $L^4 \leq N^3$ give the global bound; the additional inequality $|f_N| < \tau E_N$ gives (3.23).

Since $\mu_N \geq L/r$, the comparison balls avoid the origin. The Lipschitz bound for μ_N^{-1} gives $0.96 \leq \mu_N(x)/\mu_N(y) \leq 1.04$ on these balls. Moreover, $|\nabla \log E_N| \leq C\mu_N$ by (3.20). Integration along line segments proves the comparison for E_N. Combining this with the frequency comparison gives the estimate for the second weight. □

4 Uniform density of the small-value set

The clamped correction requires the small-value set to occupy a small fraction of every ball at the local frequency scale. The density threshold must be chosen uniformly before ε, L are fixed. We prove this by compactness of the normalized profiles. The only degenerate regime is the coalescence of the two angular modes, which is described by the following lemma.

Lemma 4.1 (Coalescing modes)**.** *Let $\ell_j \to \infty$, $1 \le k_j \le \ell_j$, and $a_j = k_j/\ell_j \to 0$. Suppose $x_j \in (\delta, 1/4)$, $Q_{k_j}(x_j) \to 1$ and $e^{i\psi_j} \to 1$. For arbitrary phases ϕ_j, set*

$$Z_j = 1 - Q_{k_j}(x_j)e^{i\psi_j}, \qquad b_j = (|Z_j|^2 + a_j^2)^{1/2}.$$

After passage to a subsequence, there are $\phi \in \mathbb{R}$, $c \in \mathbb{C}$, $d \ge 0$ and $q \in [16, 40]$ with

$$e^{i\phi_j} \to e^{i\phi}, \quad Z_j/b_j \to c, \quad a_j/b_j \to d, \quad Q'_{k_j}(x_j) \to q, \quad |c|^2 + d^2 = 1,$$

such that, for every fixed $m \ge 0$,

$$\frac{1 - Q_{k_j}(x_j + a_j y)e^{i(\psi_j + 2a_j s)}}{b_j} \longrightarrow c - d(qy + 2is) \quad \text{in } C^m_{\mathrm{loc}}(\mathbb{R}^2_{y,s}). \tag{4.1}$$

The real function

$$G(y, s) = e^{-y}\,\mathrm{Re}[e^{i(\phi+s)}\{c - d(qy + 2is)\}] \tag{4.2}$$

has a zero set of planar measure zero.

Proof. By (3.9), eventually $x_j \in [1/80, 1/10)$. This interval stays a fixed positive distance from the endpoints of the crossing region and lies where $\omega = 1$. Equation (3.10), together with compactness of the phases, gives the stated subsequential limits. Their normalization follows from $|Z_j/b_j|^2 + (a_j/b_j)^2 = 1$.

For a fixed compact set K and integer m, Taylor's formula gives

$$\begin{aligned} Q_{k_j}(x_j + a_j y)e^{i(\psi_j + 2a_j s)} &= e^{i\psi_j} Q_{k_j}(x_j) \\ &\quad + a_j e^{i\psi_j}(Q'_{k_j}(x_j)y + 2iQ_{k_j}(x_j)s) + R_j(y, s), \end{aligned}$$

where

$$\|R_j\|_{C^m(K)} \le C_{K,m} a_j^2. \tag{4.3}$$

To verify that the remainder bound is uniform, take two derivatives with respect to t of $Q_{k_j}(x_j + ta_j y)e^{i(\psi_j + 2ta_j s)}$. This gives

$$a_j^2 e^{i(\psi_j + 2ta_j s)}[y^2 Q''_{k_j} + 4iysQ'_{k_j} - 4s^2 Q_{k_j}](x_j + ta_j y).$$

All its fixed-order y, s derivatives are bounded by $C_{K,m}a_j^2$, since the evaluation points remain in the crossing region and the derivatives of Q_k are uniformly bounded. The integral remainder proves (4.3). Since $b_j \ge a_j$, $\|R_j/b_j\|_{C^m(K)} \le C_{K,m}a_j \to 0$. This proves (4.1).

If $d = 0$, then $|c| = 1$, so G is a nonzero sinusoid multiplied by e^{-y}. If $d > 0$, writing $c = c_1 + ic_2$ gives

$$e^y G = (c_1 - dqy)\cos(\phi + s) - (c_2 - 2ds)\sin(\phi + s).$$

For each s with $\cos(\phi + s) \ne 0$, this has at most one zero in y. The exceptional values of s form a discrete set. Fubini's theorem proves the assertion. □

Proposition 4.2 (Uniform density)**.** *For every $\nu > 0$ there is $\tau > 0$, independent of N, L, ε in Proposition 3.3, such that*

$$|D_N(\tau) \cap B_{2/\mu_N(X)}(X)| \le \nu |B_{2/\mu_N(X)}| \qquad (X \in \Omega). \tag{4.4}$$

Proof. In the core and the outer extension, $|f_N|/E_N = |\cos(m\theta)|$ and $\mu_N = m/r$, with $m = N$ or L. A ball of radius $2r/m$ has radial width at most $4r/m$ and angular width at most C/m. Its angular interval meets at most a fixed number of cosine zeros. The angular measure of the set where $|\cos(m\theta)| < \tau$ is at most $C\tau/m$ for $\tau < 1/2$. Integrating with the area element $r'\,dr'\,d\theta$ gives a relative measure at most $C\tau$. Balls meeting an assembly interface lie in matching pure-mode collars by the choice of ε_0 and satisfy the same estimate.

For any remaining ball with center X_0, use $\xi = \mu_N(X_0)(X - X_0)$ and divide f_N by $E_N(X_0)$. Equations (3.20) and (3.22) give uniform C^h bounds on a neighborhood of $\overline{B_2}$, for each fixed h. The normalized profiles are therefore precompact in every fixed C^m topology. It remains to identify their limits and show that each has a zero set of measure zero.

We may work with the exterior annular profiles. Normalization cancels the assembly coefficient up to a sign, which is fixed after passage to a subsequence. Inversion and its factor r^2 contribute a nondegenerate change of coordinates and a nonvanishing multiplier. To verify uniformity, let (r_0, θ_0) be the exterior center, set $m_0 = m(r_0)$, and write

$$y = \ell \log(r/r_0), \quad s = \ell(\theta - \theta_0), \quad x = x_0 + (k/\ell)y. \tag{4.5}$$

The Jacobians into frequency-rescaled Euclidean coordinates are

$$(m_0/\ell)^2 e^{2y/\ell} \quad \text{and, after inversion,} \quad (m_0/\ell)^2 e^{-2y/\ell}. \tag{4.6}$$

Because $1 \le m_0/\ell \le 3$, these and their reciprocals are bounded on fixed bounded coordinate sets. For bounded ℓ the maps have smoothly convergent subsequences; for $\ell \to \infty$ they converge, after a subsequence, to invertible linear maps. The normalized Kelvin multiplier is $(r_0/r)^2 = e^{-2y/\ell}$ and has a nonvanishing limit. Neither operation changes whether the zero set has measure zero.

The centers satisfy $x_0 \in [-1, 2]$. Passing to a subsequence, we may assume that x_0 and all angular phases converge. Every ball not covered by the collar argument remains within a single annular profile.

Bounded ℓ. The integers ℓ, k are fixed on a subsequence. Positivity of the envelope and compactness of the center parameters bound $E(r_0, \theta_0)/r_0^{-\ell}$ below. The limit is a radial combination of two angular waves of distinct frequencies ℓ and $\ell + 2k$. The radial coefficients never vanish simultaneously: wherever the first can vanish, the second is nonzero. Hence the angular trigonometric polynomial is nontrivial at each fixed radius. Its zero set has one-dimensional measure zero, and Fubini's theorem gives the assertion.

$\ell \to \infty$ and $k/\ell \to a_* > 0$. Here $k \to \infty$, $L_k \to 40x$ smoothly on bounded intervals, and H_k has the corresponding smooth limit. The center envelope remains bounded below after division by $r_0^{-\ell}$: wherever cancellation is possible, $\omega = 1$ and $a \to a_* > 0$. In (4.5) the limit is a nonzero radial factor times two angular waves of frequencies 1 and $1 + 2a_*$. At each radial coordinate, their coefficients are not both zero. Since the frequencies are distinct, the angular analytic function is nontrivial. Fubini's theorem again gives a zero set of planar measure zero.

$\ell \to \infty$ *and* $a = k/\ell \to 0$. Set $\phi = \ell\theta_0 \pmod{2\pi}$, $\psi_0 = 2k\theta_0 \pmod{2\pi}$ and $b = E(r_0, \theta_0)/r_0^{-\ell}$. If b stays bounded below, the radial coefficients and the relative phase become constant in the limit, while $a^2\omega^2$ tends to zero. The limit is a nonzero multiple of $e^{-y}\operatorname{Re}(e^{i(\phi+s)}Z_*)$, with $Z_* \neq 0$. If $b \to 0$, the separation of the cutoff regions forces the centers into the exactly biharmonic crossing region, and

$$Q_k(x_0) \to 1, \quad e^{i\psi_0} \to 1, \quad b = (|1 - Q_k(x_0)e^{i\psi_0}|^2 + a^2)^{1/2}.$$

Lemma 4.1 gives the limit (4.2). In either case, the zero set has planar measure zero.

Suppose (4.4) fails for a fixed $\nu > 0$. Choose $\tau_j \downarrow 0$ and corresponding balls on which the estimate fails, and rescale each ball to B_2. A subsequence of the normalized profiles converges uniformly to g with $|\{g = 0\} \cap B_2| = 0$. By envelope comparison, each rescaled small-value set is contained in $\{|g_j| < K\tau_j\} \cap B_2$. If $\delta_j = K\tau_j + \|g_j - g\|_{L^\infty(B_2)}$, then

$$\{|g_j| < K\tau_j\} \cap B_2 \subset \{|g| \le \delta_j\} \cap B_2.$$

For every $t > 0$, the upper limit of these measures is at most $|\{|g| \le t\} \cap B_2|$. Letting $t \downarrow 0$ and using continuity of measure from above gives zero, contrary to the fixed positive lower bound. The sets remain in the fixed ball B_2 throughout the argument. □

5 Clamped replacement and biharmonic averaging

The correction uses two tools: a clamped inverse estimate obtained without positivity, and a signed radial average that reproduces biharmonic functions on the sampling balls.

5.1 A planar sparse inverse

The first lemma combines a local energy estimate with a pointwise bound for functions supported on sets of small measure.

Lemma 5.1 (Local energy and small supports)**.** *Let $D \subset \mathbb{R}^2$ be open and let $w \in H_0^2(D;\mathbb{R})$ satisfy*

$$\int_D \Delta w \Delta\phi = \int_D F\phi \quad (\phi \in H_0^2(D)), \qquad F \in L^2(D;\mathbb{R}).$$

Extend w by zero. Suppose $\eta \in C_c^\infty(B_{2s}(x))$, $0 \le \eta \le 1$, $\eta = 1$ on $B_s(x)$, and $|D^j\eta| \le C_j s^{-j}$ for $j = 1, 2$. Then

$$\|D^2(\eta^2 w)\|_2^2 \le C\left(s^{-4}\int_{B_{2s}(x)} |w|^2 + \int_{D\cap B_{2s}(x)} |Fw|\right). \tag{5.1}$$

Also, if $v \in H^2(\mathbb{R}^2)$ and $|\{v \neq 0\}| \le a < \infty$, then

$$\|v\|_\infty \le Ca^{1/2}\|D^2 v\|_2. \tag{5.2}$$

The constants depend only on the indicated cutoff bounds.

Proof. Multiplication by η^4 preserves $H_0^2(D)$, by approximation. Set $v = \eta^2 w$. Expansion yields

$$|\Delta v|^2 - \Delta w \Delta(\eta^4 w) = -8\eta^2 |\nabla\eta|^2 w\Delta w + (4\eta\nabla\eta\cdot\nabla w + (2|\nabla\eta|^2 + 2\eta\Delta\eta)w)^2.$$

Integrating the first term by parts and using $|\nabla(\eta^2|\nabla\eta|^2)| \le C\eta s^{-3}$ gives

$$\|\Delta v\|_2^2 \le \int_D |F|\eta^4|w| + Cs^{-2}\int \eta^2|\nabla w|^2 + Cs^{-4}\int_{B_{2s}(x)} |w|^2. \tag{5.3}$$

All integrations are performed on the zero extensions. Approximation from $C_c^\infty(D)$ in H^2 justifies the identities; no boundary integral is discarded. Writing $G = \int \eta^2|\nabla w|^2$, a second integration by parts and substitution for $\eta^2\Delta w$ give

$$G = -\int w\Delta v + 2\int \eta w\nabla\eta\cdot\nabla w + \int (2|\nabla\eta|^2 + 2\eta\Delta\eta)w^2.$$

Hence, for $W = \|w\|_{L^2(B_{2s}(x))}$, $G \le W\|\Delta v\|_2 + G/2 + Cs^{-2}W^2$. Substituting this bound into (5.3) and applying Young's inequality allows a multiple of $\|\Delta v\|_2^2$ to be absorbed into the left-hand side. The Fourier identity $\|D^2 v\|_2^2 = \|\Delta v\|_2^2$ proves (5.1).

For (5.2), split the Fourier integral at $R > 0$. Cauchy–Schwarz gives

$$\|v\|_\infty \le C(R\|v\|_2 + R^{-1}\|D^2 v\|_2), \qquad \int_{|\xi|>R} |\xi|^{-4}\,d\xi = \pi R^{-2}.$$

Optimizing gives $\|v\|_\infty^2 \le C\|v\|_2\|D^2 v\|_2$. Using $\|v\|_2 \le a^{1/2}\|v\|_\infty$ and cancelling the nonzero factor gives the claim for nonzero v. If the Hessian norm vanishes, the L^2 function v is zero. Approximation extends the Fourier estimates to H^2 functions; the argument requires neither an endpoint embedding nor a logarithmic loss. □

We combine the small-support inequality with the local energy bound to estimate the clamped inverse on a sparse set.

Lemma 5.2 (Weighted sparse inverse)**.** *Let $\mu, A > 0$ be continuous on Ω and suppose $B_{4/\mu(x)}(x) \Subset \Omega$. Assume, for a fixed $K \ge 1$,*

$$K^{-1} \le \frac{\mu(y)}{\mu(x)} \le K, \qquad K^{-1} \le \frac{A(y)}{A(x)} \le K \quad \text{if } |y-x| < 4/\mu(x). \tag{5.4}$$

There are $\nu_0(K) > 0$ and $C(K) > 0$ with the following property. If an open set $D \subset \Omega$ satisfies

$$|D \cap B_{2/\mu(x)}(x)| \le \nu|B_{2/\mu(x)}| \quad (x \in \Omega), \qquad 0 < \nu \le \nu_0(K), \tag{5.5}$$

and a real measurable source obeys $|F| \le H\mu^4 A$ almost everywhere on D for some $H \ge 0$, then there is a real $w \in H^2_{\mathrm{loc}}(\Omega) \cap C(\Omega)$, zero on $\Omega \setminus D$, such that

$$\Delta^2 w = F \quad \text{in } \mathcal{D}'(D), \qquad |w| \le C(K)\nu^2 HA \quad \text{on } \Omega. \tag{5.6}$$

For finitely many admissible weights and source bounds, the same correction can satisfy all the corresponding estimates, after taking ν below their thresholds.

Proof. First let $D' \subset D$ be open with $\overline{D'} \Subset \Omega$. The source is bounded on D' by its weighted bound. Solve

$$\int_{D'} \Delta w \Delta \phi = \int_{D'} F\phi, \qquad w, \phi \in H_0^2(D').$$

Zero extension, the Hessian identity, and Poincaré's inequality on a ball containing D' imply coercivity on $H_0^2(D')$. Lax–Milgram therefore gives a solution. By the planar H^2 embedding, its zero extension is continuous and has compact support in Ω. Thus $S = \sup_\Omega |w|/A$ is finite and, if positive, attained at some x. Set $s = \mu(x)^{-1}$. On $B_{2s}(x)$,

$$|w| \le KSA(x), \qquad |F| \le K^5 H s^{-4} A(x) \quad \text{on } D'.$$

The integrals involving w have support of measure at most $C\nu s^2$. Lemma 5.1 gives

$$\|D^2(\eta^2 w)\|_2^2 \le C(K)\nu s^{-2} A(x)^2 (S^2 + HS).$$

Applying (5.2) to $\eta^2 w$ then gives

$$S^2 A(x)^2 \le C(K)\nu^2 A(x)^2 (S^2 + HS).$$

Choose ν_0 so that $C(K)\nu_0^2 \le 1/2$. Absorbing the quadratic term and cancelling the remaining positive factor gives $S \le 2C(K)\nu^2 H$. The estimate also holds when $S = 0$.

To remove the compactness restriction, take $D_j = D \cap \{j^{-1} < |x| < j\}$. The corresponding variational solutions satisfy the pointwise bound uniformly. Equation (5.1) on fixed compact subsets of Ω, together with the pointwise bound, gives uniform local H^2 bounds. A diagonal subsequence converges weakly in H^2_{loc} and locally uniformly, by compact embedding into $C^{0,\beta}$ on smaller balls for $0 < \beta < 1$. The limit vanishes on $\Omega \setminus D$ and inherits the pointwise bound. Every compactly supported test function in D is admissible in D_j for large j, so weak convergence proves the equation. Every approximant satisfies all the finitely many weighted estimates, so the same limit inherits them simultaneously. We do not assert the global identity $\Delta^2 w = F\mathbf{1}_D$ for the zero extension. □

5.2 Averaging with a vanishing second moment

We smooth the correction across the boundary of the small-value set by an average that reproduces biharmonic functions. Choose a real radial $\psi \in C_c^\infty(B_1^2)$ such that

$$\int \psi = 1, \qquad \int |z|^2 \psi(z)\, dz = 0. \tag{5.7}$$

For example, if ρ is a nonnegative radial smooth unit-mass kernel, take $\psi = (4\rho_{1/2} - \rho)/3$, where $\rho_b(z) = b^{-2}\rho(z/b)$. This kernel is signed. For a positive smooth radius $a(x)$, put

$$\mathcal{S}_a h(x) = \int \psi(z) h(x + a(x)z)\, dz.$$

Lemma 5.3 (Biharmonic averaging)**.** *Assume the sampling balls lie in the domain of the functions and $|D^j a| \le C_j a^{1-j}$ for $1 \le j \le 4$. For locally bounded h,*

$$|D^j \mathcal{S}_a h(x)| \le C a(x)^{-j} \sup_{B_{a(x)}(x)} |h|, \qquad 0 \le j \le 4. \tag{5.8}$$

For smooth f,

$$|\Delta^2 \mathcal{S}_a f(x)| \le C \sup_{B_{a(x)}(x)} |\Delta^2 f|. \tag{5.9}$$

If h is distributionally biharmonic on the sampling ball, then $\mathcal{S}_a h(x) = h(x)$ for its smooth representative. The constants depend only on ψ and the radius derivative bounds. Bounds of the same type at all orders give smoothness of $\mathcal{S}_a h$.

Proof. Write the averaging kernel as $G(x,y) = a(x)^{-2}\psi((y-x)/a(x))$. The chain rule gives $\|D_x^j G(x,\cdot)\|_1 \le C a(x)^{-j}$. On each compact subset, the kernels and their derivatives have a common compact support within the domain. Differentiation under the integral is therefore justified for locally bounded h.

For a smooth biharmonic h, its circular mean $M_h(r)$ satisfies

$$M_h'' + r^{-1} M_h' = \Delta h(x), \qquad M_h(0) = h(x), \quad M_h'(0) = 0,$$

since Δh is harmonic. Thus $M_h(r) = h(x) + r^2 \Delta h(x)/4$ exactly. The moment conditions (5.7) therefore give exact reproduction. In the distributional case, interior regularity for constant-coefficient elliptic operators supplies the smooth representative.

To prove (5.9), let

$$\Phi(y) = \frac{1}{8\pi} |y|^2 \log |y|, \quad T_b = \Phi * \psi_b - \Phi, \quad \psi_b(y) = b^{-2}\psi(y/b).$$

Then $\Delta^2 \Phi = \delta_0$. Choose $0 < \kappa < 1$ with $\operatorname{supp}\psi \subset B_\kappa$. Reproduction away from the singularity gives $\operatorname{supp} T_b \subset \overline{B_{\kappa b}} \subset B_b$. Also

$$\Phi(bY) = b^2 \Phi(Y) + \frac{b^2 \log b}{8\pi} |Y|^2.$$

The additional quadratic term cancels: radiality and (5.7) give

$$\int |Y - z|^2 \psi(z)\, dz - |Y|^2 = -2Y \cdot \int z\psi(z)\, dz + \int |z|^2 \psi(z)\, dz = 0.$$

Consequently $T_b(y) = b^2 T_1(y/b)$. For $q \ge 1$, $\partial_b^q T_b$ is smooth since the singular term $-\Phi$ does not depend on b. Scaling gives

$$\|\partial_b^q D_y^\beta T_b\|_1 \le C_{q,\beta} b^{4-q-|\beta|}, \qquad q \ge 1, \quad q + |\beta| \le 4. \tag{5.10}$$

Set $F = \Delta^2 f$. Compact support and distributional integration by parts give the local identity

$$\mathcal{S}_a f(x) - f(x) = \int T_{a(x)}(x - y) F(y)\, dy.$$

Differentiate with y fixed, so that no derivative falls on F. When Δ_x^2 is applied, the terms in which the radius is not differentiated combine *distributionally* to $(\psi_{a(x)} * F)(x) - F(x)$. Thus no L^1 bound is needed for the individual fourth derivatives of the kernel. Every other term contains $\partial_b^q D_y^\beta T_b$, $q \geq 1$, and a product $\prod_{h=1}^q D^{\gamma_h} a$ with $|\gamma_h| \geq 1$ and $\sum_h |\gamma_h| = 4 - |\beta|$. The product is bounded by $Ca^{q+|\beta|-4}$, exactly cancelling the power in (5.10). Adding $\Delta^2 f = F$ proves the estimate. Since all kernels are supported in the sampling ball, the argument is local and requires no extension beyond the domain. □

6 Completion of the real construction

The same variational correction will satisfy two weighted bounds: one controls its size relative to the small-value threshold; the other controls the smoothed residual and preserves the leading term.

6.1 Parameter choices and the two-weight correction

Fix all cutoff profiles, the averaging kernel, and the uniform constants in Propositions 3.3 and 4.2. Choose $\nu > 0$ below the sparse-inverse thresholds for E_N and $E_N\mu_N^{-4}$, and small enough that $C_2\nu^2 \leq 1/8$ in (6.4) below. The constant C_2 depends only on those uniform constants and $\|\psi\|_1$. Next choose $0 < \tau < 1/10$ from Proposition 4.2. Fix $0 < \varepsilon \leq \varepsilon_0$ with $\varepsilon^2 \leq \tau$ and an even $L \geq \max\{L_0, 16, 4\varepsilon^{-2}\}$. The smoothing parameter σ will be fixed below in terms of these constants, and then N_0 is chosen so that $N > L$ and $N^3 \geq L^4$ for every even $N \geq N_0$. Thus the order is

$$\text{fixed profiles} \longrightarrow \nu \longrightarrow \tau \longrightarrow \varepsilon \longrightarrow L \longrightarrow \sigma \longrightarrow N_0,$$

and only N subsequently varies.

Apply Lemma 5.2 in $D_N = D_N(\tau)$ to $F = -\Delta^2 f_N$. With weight E_N, (3.23) gives $H \leq C(\varepsilon^2 + \tau) \leq C\tau$. With weight $E_N\mu_N^{-4}$, (3.21) gives $H = M_N$. Using the same variational approximants for both weights gives a single real correction $w_N \in H^2_{\mathrm{loc}}(\Omega) \cap C(\Omega)$, zero off D_N, satisfying

$$\Delta^2(f_N + w_N) = 0 \quad \text{in } D_N,$$

$$|w_N| \leq C\nu^2\tau E_N, \tag{6.1}$$

$$|w_N| \leq C\nu^2 M_N \mu_N^{-4} E_N. \tag{6.2}$$

The corrected function is smooth and biharmonic in D_N. Only this interior equation is used; the argument does not require a global inhomogeneous equation for the zero extension.

Set

$$a_N(x) = \sigma/\mu_N(x), \qquad u_N^\circ = \mathcal{S}_{a_N}(f_N + w_N). \tag{6.3}$$

Take $0 < \sigma \leq 1$. Equation (3.19) implies $|D^j a_N| \leq C_{j,\sigma} a_N^{1-j}$ at every fixed order, uniformly in N. For example $|D^j \mu_N^{-1}| \leq C_j \mu_N^{j-1}$ follows by differentiating the reciprocal. All sampling balls avoid zero, and u_N° is smooth on Ω.

6.2 The dichotomy and the bounded real potential

We separate the region where averaging reproduces the biharmonic replacement from the region where a lower bound controls the quotient. The gradient estimate for f_N, local comparison, and (6.1) give

$$|u_N^\circ - f_N| \le C_1\sigma E_N + C_2\nu^2\tau E_N. \tag{6.4}$$

To see this, write $\mathcal{S}_{a_N} f_N - f_N$ using $\int \psi = 1$ and integrate the gradient along each sampling segment. Estimate $\mathcal{S}_{a_N} w_N$ by $\|\psi\|_1$. The constants C_1, C_2 are therefore independent of $\sigma, \tau, \varepsilon, L, N$ for $\sigma \le 1$; this step uses no derivatives of the averaging kernel. Choose σ so that $C_1\sigma \le \tau/8$. Then

$$|u_N^\circ - f_N| \le \tau E_N/4. \tag{6.5}$$

Decrease the fixed ratio σ/τ further so that on every sampling ball

$$E_N(y) \ge \tfrac{7}{8} E_N(x), \qquad |f_N(y) - f_N(x)| \le \tfrac{1}{8}\tau E_N(x).$$

These follow from (3.20) and comparison. If $|f_N(x)| < \tau E_N(x)/2$, then throughout the ball

$$|f_N(y)| < \tfrac{5}{8}\tau E_N(x) < \tau E_N(y),$$

so the sampling ball lies strictly inside D_N. Biharmonic reproduction gives $u_N^\circ = f_N + w_N$ on the open set $\{|f_N| < \tau E_N/2\}$. We obtain the dichotomy

$$\begin{cases} \Delta^2 u_N^\circ = 0, & |f_N| < \tau E_N/2, \\ |u_N^\circ| \ge \tau E_N/4, & |f_N| \ge \tau E_N/2. \end{cases} \tag{6.6}$$

In particular, the function is biharmonic on an open neighborhood of every zero in Ω.

By Lemma 5.3 and (6.2),

$$\begin{aligned} |\Delta^2 u_N^\circ(x)| &\le C \sup_{B_{a_N(x)}(x)} |\Delta^2 f_N| + C a_N(x)^{-4} \sup_{B_{a_N(x)}(x)} |w_N| \\ &\le C_{\sigma,\nu} M_N E_N(x). \end{aligned}$$

Define on Ω

$$V_N(x) = \begin{cases} -\Delta^2 u_N^\circ(x)/u_N^\circ(x), & u_N^\circ(x) \ne 0, \\ 0, & u_N^\circ(x) = 0. \end{cases} \tag{6.7}$$

On the first set in (6.6) it is identically zero; on the second, $|V_N| \le C_{\sigma,\nu} M_N/\tau \le CN^3$. All auxiliary parameters are already fixed, so C is independent of N. The potential is real and smooth on Ω. Where u_N° is nonzero, it is a quotient of smooth functions; near every zero, it vanishes identically on an open neighborhood.

6.3 The leading term and regularity at zero

In the fixed core, (3.15) and (3.18) give $\mu_N = N/r$ and $E_N = |c_N| r^{N+2}$. For sufficiently small r the sampling ball stays in the core. The core polynomial is biharmonic, so $\mathcal{S}_{a_N} f_N = f_N$ there. The second correction estimate, combined with differentiation of the averaging kernel, gives

$$|D^j \mathcal{S}_{a_N} w_N(x)| \le C_{N,j} |x|^{N+6-j}, \qquad 0 \le j \le 4. \tag{6.8}$$

The undifferentiated bound is $C\nu^2 M_N N^{-4} |c_N| r^{N+6}$, and the jth derivative introduces the factor $a_N^{-j} = (N/(\sigma r))^j$. Local comparison controls the supremum on the sampling ball. Dependence on N is allowed in (6.8).

Let $R_N = \mathcal{S}_{a_N} w_N$ near zero. Its derivatives through order four tend to zero. For $|\beta| \le 3$,

$$\frac{|D^\beta R_N(he_i)|}{|h|} \le C_{N,\beta} |h|^{N+5-|\beta|} \longrightarrow 0.$$

Induction on $|\beta|$, followed by continuity at order four, shows that the extended derivatives are the classical derivatives of the zero extension of the remainder. Thus $u_N^\circ \in C^4(\mathbb{R}^2)$ and

$$u_N^\circ = c_N |z|^2 \operatorname{Re}(z^N) + O_N(|z|^{N+6})$$

with the derivative bounds in (6.8). The supremum on a small ball is at most $C_N r^{N+2}$. Evaluation at $(r/2, 0)$ gives the lower bound $c'_N r^{N+2}$ for sufficiently small r, because $c_N \ne 0$. Hence $\operatorname{ord}_0 u_N^\circ = N + 2$.

Set $V_N(0) = 0$. Both $u_N^\circ(0)$ and $\Delta^2 u_N^\circ(0)$ vanish: the leading polynomial is biharmonic and the fourth derivatives of the remainder tend to zero. The equation holds pointwise everywhere. The distributional fourth derivatives of a C^4 function agree with its classical derivatives, so no residual is supported at the origin.

6.4 Fixed normalization

At $(1/2, 0)$, $f_N = E_N = 1/4$, whereas for $r \ge 1$, $E_N = 2^{-L-2} r^{-L}$. Hence (6.5) gives

$$|u_N^\circ(1/2, 0)| \ge \tfrac{1}{4}(1 - \tau/4), \qquad \sup_{|x| \ge 1} |u_N^\circ(x)| \le (1 + \tau/4) 2^{-L-2}.$$

The lower bound at the interior point is strictly larger than the upper bound outside the unit ball. Since L is fixed, the envelope is bounded; moreover, u_N° is continuous and tends to zero at infinity. Thus

$$K_N = \max_{x \in \mathbb{R}^2} |u_N^\circ(x)| > 0$$

is finite and attained inside B_1. Define $u_N = u_N^\circ / K_N$. The potential and the order are unchanged, (1.6) holds, and (1.7) follows with $d_N = c_N / K_N \ne 0$. This proves Theorem 1.2.

7 The upper estimate and sharpness in all dimensions

We prove the normalized upper bound independently of the construction, then use constant extension to obtain sharpness in all three classes.

7.1 Carleman input and normalization

For [13], we use the numbering and pagination of arXiv:1505.05228v1. Lemma 2.1, p. 3, with $m = 2$, gives a positive nondecreasing weight $\varpi \asymp r$ on $(0, 10)$ and

$$\lambda^6 \int \varpi(|x|)^{-1-2\lambda}|h|^2 \le C(n) \int \varpi(|x|)^{5-2\lambda}|\Delta^2 h|^2, \tag{7.1}$$

for $h \in C_c^\infty(B_{10} \setminus \{0\})$ and $\lambda \ge \lambda_0(n)$. Lemma 2.2, pp. 3–4, applied to the homogeneous elliptic operator Δ^2 and rescaled to a ball, gives

$$\|D^j u\|_{L^\infty(B_{s/2}(y))} \le C(n)s^{-j}(1 + Ms^4)\|u\|_{L^\infty(B_s(y))}, \qquad 0 \le j \le 3, \tag{7.2}$$

when $\Delta^2 u = -Vu$ and $\|V\|_\infty \le M$ there. Only boundedness of the right-hand side is used. For complex u, V, apply the interior estimate to the real and imaginary parts of u; their right-hand sides have modulus at most $M\|u\|_\infty$. Adding the squared componentwise Carleman estimates gives the complex version. No derivative of V is required.

We first arrange a strict gap between the two fixed weight values:

$$\varpi(21/10) < \varpi(3). \tag{7.3}$$

Otherwise, if $\varpi(21b/10) = \varpi(3b)$ for every $b \in (0, 1]$, iteration would give $\varpi(t) = \varpi(0.7^j t)$ for $0 < t \le 3$, contrary to $\varpi(r) \asymp r$. For a fixed b with strict inequality, replace $\varpi(r)$ by $b^{-1}\varpi(br)$. Applying (7.1) to $h(\cdot/b)$ shows that the same estimate holds with the constant multiplied by b^{-2}, independently of M, λ. We use the rescaled weight below.

Proof of Proposition 1.3. Fix $p \in B_1$ and $0 < r < 1/2$. Take an outer cutoff equal to one on $B_3(p)$ and supported in $B_4(p)$, and an inner cutoff zero on $B_{r/4}(p)$ and one outside $B_{r/2}(p)$. Let their product be χ. Inner derivatives of order j cost $C_j r^{-j}$; outer derivatives have fixed bounds. The support lies in $B_5 \Subset B_{10}$. For fixed r, approximate χu in H^4 by smooth functions with a common compact support away from p. The weights are bounded on this support, so both weighted integrals converge. Consequently, (7.1) applies to χu with the weight centered at p.

Put $S = \|u\|_{L^\infty(B_r(p))}$. If $S \ge 1/2$, the claim follows by decreasing c. Otherwise choose $z \in B_1$ with $|u(z)| \ge 3/4$. The interior estimate gives a Lipschitz representative on B_6 with Lipschitz constant $A = C(n, C_0)(1 + M) \ge 1$. For $y \in B_r(p)$, $A|z - y| \ge |u(z)| - |u(y)| \ge 1/4$. With $d = (16A)^{-1}$, $B_d(z)$ is outside $B_r(p)$, inside $B_{21/10}(p)$, and satisfies $|u| \ge 1/2$ and $\chi = 1$. Therefore, with $\varpi = \varpi(|x - p|)$,

$$J := \int \varpi^{-1-2\lambda}|\chi u|^2 \ge c(n, C_0)(1 + M)^{-n}\varpi(21/10)^{-1-2\lambda}. \tag{7.4}$$

Take $\lambda \ge \max\{\lambda_0(n), n + 3\}$. The potential contribution in (7.1) is at most $CM^2 \sup_{\operatorname{supp}\chi} \varpi^6$ times J, so it is absorbed for $\lambda \ge C(n)(1 + M^{1/3})$. The commutator is

$$[\Delta^2, \chi]u = 4\nabla\chi \cdot \nabla\Delta u + 2\Delta\chi\Delta u + 4\sum_{i,j} \partial_{ij}\chi\partial_{ij}u$$
$$+ 4\nabla\Delta\chi \cdot \nabla u + (\Delta^2\chi)u.$$

On the inner annulus, use (7.2) on balls of radius comparable to r contained in $B_r(p)$. The commutator is bounded by $Cr^{-4}(1 + Mr^4)S$. Its squared weighted integral satisfies

$$I_{\text{in}} \le C(n)^{\lambda+1} r^{n-3-2\lambda}(1 + M)^2 S^2.$$

On the outer annulus, fixed-radius interior estimates and $5 - 2\lambda < 0$ give

$$I_{\text{out}} \le C(n, C_0)(1 + M)^2 \varpi(3)^{5-2\lambda}.$$

Set $\theta = \varpi(21/10)/\varpi(3) < 1$. Dividing the last bound by λ^6 times (7.4) gives at most

$$C(n, C_0)(1 + M)^{n+2} \lambda^{-6} \theta^{2\lambda}.$$

Since $\log(1 + M) \le CM^{1/3}$ for $M \ge 1$, a fixed choice $\lambda = T(n, C_0)(1 + M^{1/3})$ absorbs both the outer contribution and the potential term. Incorporating the fixed weight values into the exponential constants, the remaining inequality gives

$$\begin{aligned} S &\ge c(n, C_0)C(n)^{-\lambda}(1 + M)^{-(n+2)/2} r^{\lambda+(3-n)/2} \\ &\ge c(n, C_0) e^{-D(n,C_0)(1+M^{1/3})} r^{\lambda+1/2}. \end{aligned}$$

The second inequality uses $n \ge 2$ and $r < 1$. For $r < 1/4$, the exponential factor is at least $r^{(D/\log 4)(1+M^{1/3})}$, proving (1.8). A vanishing order strictly larger than the resulting exponent would contradict this lower bound as $r \downarrow 0$. □

7.2 Extension and the three classes

We record the preservation of essential suprema and vanishing order under constant extension.

Lemma 7.1. *Let $2 \le d \le n$, $u \in H^4_{\text{loc}}(B^d_{10}) \cap L^\infty(B^d_{10})$, and $V \in L^\infty(B^d_{10})$ solve $\Delta_d^2 u + Vu = 0$ distributionally. For $x \in \mathbb{R}^d$, $y \in \mathbb{R}^{n-d}$ define $U(x, y) = u(x)$ and $W(x, y) = V(x)$. Then*

$$\Delta_n^2 U + WU = 0 \quad \text{in } B^n_{10},$$

and, for $0 < r \le 10$,

$$\|U\|_{L^\infty(B^n_r)} = \|u\|_{L^\infty(B^d_r)}, \qquad \|W\|_{L^\infty(B^n_r)} = \|V\|_{L^\infty(B^d_r)}. \tag{7.5}$$

In particular, $\operatorname{ord}_0 U = \operatorname{ord}_0 u$.

Proof. There is nothing to extend when $n = d$. Otherwise, all derivatives in y vanish. For weak derivatives, this follows from Fubini's theorem applied to compactly supported test functions. On compact cylinders, the derivatives in x through order four belong to L^2. Hence $U \in H^4_{\text{loc}}$, and the equation holds distributionally. Projection of B^n_r gives one inequality in each norm identity. For the reverse inequality, take $0 < t < r$ and use the positive-volume cylinder

$$B^d_t \times B^{n-d}_{\sqrt{r^2-t^2}} \subset B^n_r.$$

Fubini identifies its essential supremum with that on B^d_t. Letting $t \uparrow r$ proves (7.5), and then the definition of order gives the last assertion. □

Proof of Corollary 1.4. Extend the real pair of Theorem 1.2 constantly in the last $n-2$ coordinates. Lemma 7.1 preserves the equation, the normalization with $C_0 = 1$, the potential bound, and the order $N+2$. The extension is C^4, hence in H^4_{loc}.

By Proposition 1.3, the exponent $1/3$ is admissible in the largest class $\mathcal{A}_n^{\mathbb{C}}$. If $\alpha < 1/3$ were admissible in the smallest class $\mathcal{A}_n^{\mathbb{R}}$, apply (1.4) to this real family with $M = \max\{1, \|V_N\|_\infty\}$ and $p = 0$. Since $\|V_N\|_\infty \le CN^3$ by Theorem 1.2, it would give

$$N + 2 \le C'(1 + M^\alpha) \le C''(1 + N^{3\alpha})$$

for all sufficiently large even N, a contradiction. Since

$$\mathcal{A}_n^{\mathbb{R}}(C_0, M) \subset \mathcal{A}_n^{\mathbb{R},\mathbb{C}}(C_0, M) \subset \mathcal{A}_n^{\mathbb{C}}(C_0, M),$$

all three infima equal $1/3$, with the asserted endpoint bound. No separate complex-valued example is needed. □

8 Concluding remarks

Remark 8.1 (Real zeros and the two weights). Restricting to real solutions and real potentials does not improve the upper-bound exponent. The additional difficulty in the construction is the larger nodal set of a real angular wave. The positive envelope regularizes amplitude cancellation, and the sparse inverse removes the residual near the remaining real zeros. The two estimates for w_N serve different purposes: the first gives a lower bound for the smoothed function outside the smaller small-value region; the second preserves the prescribed polynomial jet and controls the potential after four derivatives fall on the averaging kernel.

Remark 8.2 (Scope of the construction). The potentials are real and have no prescribed sign. No uniform Hölder or derivative bound, compact support, or smoothness at the origin is claimed. After extension to $\mathbb{R}^n$, smoothness is guaranteed off the axis $\{x_1 = x_2 = 0\}$, not necessarily off the single origin. The zeros need not be isolated and no boundary condition is imposed. Applying the independently proved upper estimate with the actual norms of the constructed planar potentials also gives $cN^3 \le \|V_N\|_\infty \le CN^3$ for all sufficiently large N. This consequence is not used in the construction.

Remark 8.3 (Local order versus decay at infinity). Under $u_R(x) = u(x_0 + Rx)$ the potential becomes $V_R(x) = R^4 V(x_0 + Rx)$, so the exponent $1/3$ corresponds algebraically to $R^{4/3}$. The finite construction here establishes sharpness for a normalized local vanishing-order bound. It does not by itself produce a single solution with prescribed exponential decay in every direction at infinity, nor does it establish the optimality of a logarithmic factor in a Landis-type estimate. These conclusions require additional global arguments.

Remark 8.4 (Dimension one). For $u^{(4)} + Vu = 0$ on an interval, with $V \in L^\infty_{\mathrm{loc}}$ and $u \in H^4_{\mathrm{loc}}$, the vector $(u, u', u'', u^{(3)})$ is locally absolutely continuous and solves a first-order system with locally bounded coefficients. Gronwall's inequality gives uniqueness for zero initial data. A nontrivial solution therefore has a nonzero derivative of order at most three at each point, so its vanishing order is at most three. The analogous optimal power is zero in dimension one.

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